TIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The 12:1 Epicyclic Reduction

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why do ordinary toy clocks slip and break when children turn them backwards?'
- Ask: 'How does the planetary gear core prevent fingers from getting pinched?'
- Ask: 'Why is there 360 degrees in a full circle and 30 degrees between each hour?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students trying to force the hour pointer directly with their thumbs. Direct them to drive the clock smoothly using the minute hand drive shaft.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"If the minute hand turns through 90° (15 minutes), how many degrees does the hour hand move?"
""

OFFICIAL SOLUTION KEY:

Hour hand moves 30° / 4 = 7.5°.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

TIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The 12:1 Epicyclic Reduction

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does turning the long hand move the short hand automatically? Inside this clock is a miniature planetary gearbox identical to the transmissions in spacecraf

STEP-BY-STEP APPARATUS SETUP:

1. Hold the clock body in your hands.
2. Align both the hour pointer and minute pointer to 12:00.
3. Slowly rotate the minute hand one full revolution clockwise.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Planetary Clock Core (Herringbone Gearbox) + Detachable Dial Face

DATUM TARGET: Gear Invariant: 1 Full Turn of Minute Hand (360°) -> Exactly 30° Hour Hand Advance

CALCULATOR KEY: [CLK] [1] [:] [00] [+] [1] [:] [00] -> 2:00

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 1B, Chapter 15 (Telling Time to the Hour)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 1, Unit 4 (Clocks & Hours)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 1, Ch. 9 • CCSS.MATH.1.MD.B.3 • TEKS 1.7E

GRADE LEVEL BAND: Grades 1–3 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 6 ('Geometry & Time'), pp. 128–140

HARDWARE UNBOXING BOOKLET: TIME-01 Booklet: Page 2 (12:1 Epicyclic Ratio) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE 12:1 REDUCTION CHECK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Watch the hour pointer as the minute hand completes 1 full turn (360°).

Step 2: Notice: The hour hand advances from 12 to 1 (exactly 30° arc, which is 1/12 of 360°).

Step 3: Rotate the minute hand another full turn. Notice: Hour pointer moves exactly from 1 to 2.

Step 4: Conclude: 1 hour = 60 minutes. The 12:1 gear ratio guarantees that the hands can NEVER slip out of synchronization!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Gear Invariant: 1 Full Turn of Minute Hand (360°) -> Exactly 30° Hour Hand Advance

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core (Herringbone Gearbox) + Detachable Dial Face

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: REVERSE ROTATION AUDIT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Rotate the minute hand counter-clockwise by 2 full turns.

FORM A MASTER STATION TASKS (Complete All 3):

1. **What time does the clock display after 2 reverse turns from 2:00?**
   Expected Physical Finding: **12:00**

2. **Did the gears slip or jam during reverse motion?**
   Expected Physical Finding: **Zero slip (Herringbone captive teeth)**

3. **How many minute turns are needed to move the hour hand 180° (6 hours)?**
   Expected Physical Finding: **6 full revolutions**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [CLK] [1] [:] [00] [+] [1] [:] [00] -> 2:00

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1:00	X: 1:00	Initial time state
ENTER	Y: 1:00 | X: 1:00	Stack lift
1:00	Y: 1:00 | X: 1:00	Interval duration
+	X: 2:00	Time addition evaluates new hour

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

TIME-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 4: Halfway There: Continuous Motion

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Where would the hour hand point at 8:30?'
- Ask: 'Can the hour pointer ever point directly at a number when the minute hand is at 6?'
- Ask: 'How does the gear drive keep you from being fooled on tests?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students reading 5:50 as 6:50. Have them trace the arc backwards to 5:00 to see that the hour hand has not cro
ssed the 6:00 finish line.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"At what exact time between 1:00 and 2:00 are the hour and minute hands pointing in EXACTLY opposite directions (180"
"°)?"

OFFICIAL SOLUTION KEY:

At approximately 1:38 and 11 seconds (1:38.18).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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within the purchasing teacher's classroom or homeschool household.
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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 1: Halfway There: Continuous Motion

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

The #1 mistake kids make on tests is thinking that at 3:30, the hour hand still points directly at 3. Watch how physical kinematics kills this mistake forever.

STEP-BY-STEP APPARATUS SETUP:

1. Set the clock to 3:00 (minute hand at 12, hour pointer at 3).
2. Slowly turn the minute hand half a turn until it points straight down at 6 (30 minutes).
3. Look closely at the red hour pointer: Where is it pointing?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Planetary Clock Core + Detachable Half-Hour Overlay Disc

DATUM TARGET: Kinematic Invariant: At :30 minutes, Hour Pointer is EXACTLY 15° (midway between numbers)

CALCULATOR KEY: [3] [:] [30] -> Display: 3:30 (Hour hand at 3.5)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2B, Chapter 10 (Telling Time to 5 Minutes)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 5 (Clocks & Half-Hours)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 12 • CCSS.MATH.2.MD.C.7 • TEKS 2.9G
GRADE LEVEL BAND:	Grades 1–3 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 132–136
HARDWARE UNBOXING BOOKLET:	TIME-01 Booklet: Page 2 (Continuous Motion) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE MIDWAY HORIZON

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Notice: The hour pointer has crawled forward. It does NOT point at 3!

Step 2: Lay a straightedge from the center arbor through the hour pointer: It points EXACTLY halfway between 3 and 4 (at the 15° tick).

Step 3: Now advance to 3:45 (minute hand at 9): Notice the hour pointer is 3/4 of the way to 4.

Step 4: Conclude: The hour hand NEVER jumps suddenly at the bell—it glides continuously at 1/12 the speed of the minute hand.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Kinematic Invariant: At :30 minutes, Hour Pointer is EXACTLY 15° (midway between numbers)

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core + Detachable Half-Hour Overlay Disc

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE 5:50 NEAR-MISS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Set the clock to 5:50 (minute hand at 10). Where is the hour pointer?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Is the hour pointer closer to 5 or closer to 6?**
   Expected Physical Finding: **Much closer to 6 (it has traveled 50/60 = 83% of the way)**

2. **Why do many students mistakenly read 5:50 as 6:50?**
   Expected Physical Finding: **Because the hour hand looks like it is pointing at 6**

3. **How do you know it is still 5:00 and not 6:00?**
   Expected Physical Finding: **The minute hand has not yet reached 12**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [3] [:] [30] -> Display: 3:30 (Hour hand at 3.5)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3:00	X: 3:00	Base hour
30m	Y: 3:00 | X: 0:30	Elapsed interval
+	X: 3:30	Hour hand position = 3 + 30/60 = 3.5

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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TIME-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 4: Concentric Track Reading

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why are there 12 numbers on the clock if there are 60 minutes in an hour?'
- Ask: 'How does multiplying by 5 help you read the minute hand faster?'
- Ask: 'What track do you look at to read 4:08 vs 4:40?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students confusing 4:08 with 4:40. Point out that the minute hand points at 8 for 4:40, but at 1 + 3 ticks for 4:08.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A clock shows 9:47. How many minutes remain until 10:00?"
""

OFFICIAL SOLUTION KEY:

60 − 47 = 13 minutes.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

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- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 1: Concentric Track Reading

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How does a clock dial pack three different measuring tracks into a single circular face? In this lab, you master the 3 concentric reading tracks.

STEP-BY-STEP APPARATUS SETUP:

1. Inspect the detachable dial face.
2. Trace the three concentric circular paths from outside to inside.
3. Track 1 (Outer): 60 micro-ticks (1 minute each). Track 2 (Middle): 12 five-minute intervals. Track 3 (Inner): 12 hour numerals.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Planetary Clock Core + Standard 3-Track Detachable Dial Face
DATUM TARGET:	Concentric Geometry: Outer Rim = 60 Ticks (1 min) | Mid = 12 Intervals (5 min) | Inner = 12 Hours
CALCULATOR KEY:	[5] [ENTER] [7] [*] -> 35 Min (Hand at 7)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2B, Chapter 10, Lesson 10.2 (Minutes Past the Hour)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 5, Module 2 (Skip Counting Minutes by 5s)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 12, Sec. 12.2 • CCSS.MATH.2.MD.C.7 • TEKS 2.9G
GRADE LEVEL BAND:	Grades 2–4 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 136–140
HARDWARE UNBOXING BOOKLET:	TIME-01 Booklet: Page 5 (Dial Features & Tracks)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 2: LAB INVESTIGATION: THE 5-MINUTE SKIP COUNT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Point the minute hand at numeral 1: Read the middle track: '5 min'.

Step 2: Advance to 2: '10 min'. Advance to 3: '15 min'.

Step 3: Skip count by 5s around the dial: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60.

Step 4: Point the minute hand at 8: Calculate 8 × 5 = 40 minutes. Confirm with the outer tick marks.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Concentric Geometry: Outer Rim = 60 Ticks (1 min) | Mid = 12 Intervals (5 min) | Inner = 12 Hours

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core + Standard 3-Track Detachable Dial Face

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 2–4

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: READING TO THE EXACT MINUTE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Set the clock to 7:23. Use all three concentric tracks.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Where does the hour pointer sit?**
   Expected Physical Finding: **Between 7 and 8 (about 1/3 of the way)**

2. **How do you find 23 on the minute track?**
   Expected Physical Finding: **Find 4 (20 min) on mid track, then count 3 ticks past it on outer track**

3. **Write the time in standard digital format.**
   Expected Physical Finding: **7:23**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [5] [ENTER] [7] [*] -> 35 Min (Hand at 7)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
7	X: 7	Hour numeral on dial
5	Y: 7 | X: 5	Minutes per dial sector
*	X: 35	Calculate minute count via skip-counting

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 2–4

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TIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: Elapsed Time as Angular Arcs

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can\'t you just subtract 2:45 – 1:15 like ordinary decimals?'
- Ask: 'How does breaking elapsed time into landmark hours (like 11:00 or 12:00) make it easy?'
- Ask: 'How many 15-minute sector blocks fit around the entire clock face?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students subtracting 10:45 from 12:15 using base-10 borrowing (getting 1.70 instead of 1 hr 30 min). Remind th
em: time resets at 60, not 100!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Flight 412 departs at 8:50 PM and lands at 11:25 PM. What is the flight duration?"
""

OFFICIAL SOLUTION KEY:

8:50 to 9:00 = 10 min. 9:00 to 11:00 = 2 hr. 11:00 to 11:25 = 25 min. Total = 2 hr 35 min.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: Elapsed Time as Angular Arcs

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How long did soccer practice last? Instead of doing confusing base-60 math in your head, measure elapsed time as a physical angular wedge.

STEP-BY-STEP APPARATUS SETUP:

1. Dock the clock stand in the center of the table.
2. Problem: A movie starts at 1:15 and ends at 2:45. How long was the movie?
3. Set the clock to 1:15.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Clock Core + 15-Minute Arc Blocks (90° sectors) + Digital Practice Lid

DATUM TARGET: Angle-Time Invariant: 90° Arc = 15 Min | 180° Arc = 30 Min | 360° Arc = 60 Min

CALCULATOR KEY: [2] [:] [45] [ENTER] [1] [:] [15] [−] -> 1:30 (1 hr 30 min)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3A, Chapter 7 (Elapsed Time)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 3, Unit 5 (Time & Measurement Arcs)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 3, Ch. 12 • CCSS.MATH.3.MD.A.1 • TEKS 3.7C
GRADE LEVEL BAND:	Grades 3-5 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 140-146
HARDWARE UNBOXING BOOKLET:	TIME-01 Booklet: Page 6 (Aha! Discovery: Elapsed Arc Sums)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE SECTOR WEDGE FIT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Advance the clock from 1:15 to 2:15 (one full 360° revolution of minute hand). Record: 1 hour.

Step 2: Now advance from 2:15 to 2:45: The minute hand sweeps from 3 to 9 (a 180° semicircle arc).

Step 3: Fit two 15-minute sector blocks into this 180° gap: Exactly two 15-min blocks fit (15 + 15 = 30 minutes).

Step 4: Sum the components: 1 hour + 30 minutes = 1 hour 30 minutes (90 minutes total elapsed time).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Angle-Time Invariant: 90° Arc = 15 Min | 180° Arc = 30 Min | 360° Arc = 60 Min

PHYSICAL APPARATUS MANIFEST:

Clock Core + 15-Minute Arc Blocks (90° sectors) + Digital Practice Lid

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: BRIDGING THE HOUR BOUNDARY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

A science lab begins at 10:45 AM and ends at 12:15 PM. Find the elapsed time.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Minutes from 10:45 to 11:00 (top of the hour).**
   Expected Physical Finding: **15 minutes (one 15-min block)**

2. **Hours from 11:00 to 12:00.**
   Expected Physical Finding: **1 hour (60 minutes)**

3. **Minutes from 12:00 to 12:15.**
   Expected Physical Finding: **15 minutes**

4. **Total elapsed time.**
   Expected Physical Finding: **1 hour 30 minutes**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [2] [:] [45] [ENTER] [1] [:] [15] [−] -> 1:30 (1 hr 30 min)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
2:45	X: 2:45	End time
ENTER	Y: 2:45 | X: 2:45	Stack lift
1:15	Y: 2:45 | X: 1:15	Start time
−	X: 1:30	Elapsed interval difference: 90 minutes

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

TIME-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 4: The 24-Hour Solar Meridian

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is 24-hour time safer for hospitals and airlines than saying 8 o\'clock?'
- Ask: 'Why don\'t morning AM hours need 12 added to them?'
- Ask: 'What does AM (Ante Meridiem) and PM (Post Meridiem) mean in Latin?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding 12 to morning hours (e.g., converting 8:00 AM to 20:00). Remind them: morning is within the fi
rst 12 hours of the day!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A train leaves Paris at 22:35 and arrives in Berlin at 06:15 the next morning. How long was the journey?"
""

OFFICIAL SOLUTION KEY:

22:35 to 24:00 = 1 hr 25 min. 00:00 to 06:15 = 6 hr 15 min. Total = 7 hr 40 min.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 1: The 24-Hour Solar Meridian

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does the day reset at midnight, and why do pilots, astronauts, and doctors use 24-hour time? In this lab, flip your dial to master 24-hour military time.

STEP-BY-STEP APPARATUS SETUP:

1. Detach the standard 12-hour dial face from the clock body.
2. Flip the dial to Side B (the 24-Hour Solar Meridian Face: numbers 13 to 24 on the outer rim).
3. Snap the dial back onto the octagonal MultiBuild mounting posts.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Planetary Clock Core + Reversible 24-Hour Military Dial Face

DATUM TARGET: Modular Arithmetic: 1:00 PM = 1 + 12 = 13:00 | 11:00 PM = 11 + 12 = 23:00

CALCULATOR KEY: [13] [:] [00] [12/24] -> 1:00 PM

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 3B, Chapter 12 (24-Hour Time)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 4, Unit 5 (Solar Cycles & Navigation Time)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 4, Ch. 12 • CCSS.MATH.4.MD.A.2 • TEKS 4.8C

GRADE LEVEL BAND: Grades 3-5 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 146-152

HARDWARE UNBOXING BOOKLET: TIME-01 Booklet: Page 5 (Detachable Dual-Sided Dials)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: THE NOON CONVERSION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Set the clock to 12:00 Noon (sun at highest point in the sky).

Step 2: Advance the clock 1 hour into the afternoon: Notice the inner track reads '1', but the outer 24-hour track reads '13' (13:00).

Step 3: Advance to 6:00 PM dinner: Inner track reads '6', outer track reads '18' (18:00).

Step 4: Rule: For any PM time, add 12 to the hour: 1 + 12 = 13; 6 + 12 = 18; 11 + 12 = 23.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Modular Arithmetic: 1:00 PM = 1 + 12 = 13:00 | 11:00 PM = 11 + 12 = 23:00

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core + Reversible 24-Hour Military Dial Face

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: CONVERTING FLIGHT ITINERARIES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Convert between standard AM/PM time and 24-hour international military time.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Convert 3:45 PM to 24-hour time.**
   Expected Physical Finding: **3 + 12 = 15 -> 15:45**

2. **Convert 21:15 to standard AM/PM time.**
   Expected Physical Finding: **21 − 12 = 9 -> 9:15 PM**

3. **What is 12:00 Midnight in 24-hour time?**
   Expected Physical Finding: **00:00 (or 24:00)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: **[13] [:] [00] [12/24] -> 1:00 PM**

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
13:00	X: 13:00	Military 24-hour time
12/24	X: 1:00 PM	Toggle 12-hour AM/PM meridian display

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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TIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Digital-to-Analog Kinematic Bridge

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'How does an analog clock show you how much time is LEFT in the hour better than digital numbers?'
- Ask: 'Why does a digital clock make it harder to visualize fractions like a quarter-to or half-past?'
- Ask: 'How does the RPN calculator handle the base-60 turnover at 60 minutes?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to add 7:40 + 0:25 and getting 7:65 on ordinary calculators. The SC-6 CLK mode automatically c
onverts 65 minutes into +1 hour 5 minutes (8:05).

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A timer counts down from 45:00 minutes. After 28 minutes and 40 seconds, what time remains?"
""

OFFICIAL SOLUTION KEY:

45:00 − 28:40 = 16 minutes 20 seconds.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/classroom-clock/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

TIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Digital-to-Analog Kinematic Bridge

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why do modern smartwatches let you toggle between analog hands and digital numbers? In this culminating lab, you connect physical hand angles to digital calcula

STEP-BY-STEP APPARATUS SETUP:

1. Snap the Digital Practice Lid onto your Gridfinity clock box.
2. Slide the privacy shutter CLOSED over the digital display window.
3. Partner A sets a secret time on the planetary clock hands, then passes it to Partner B.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Planetary Clock Core + Digital Practice Lid with Sliding Shutter

DATUM TARGET: Bimodal Duality: Physical Angular Displacement (Analog) = Numerical State Vector (Digital)

CALCULATOR KEY: [CLK] [8] [:] [45] [ENTER] [30] [+] -> 9:15

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 3B, Chapter 12, Lesson 12.3 (Time Word Problems)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 4, Unit 5 (Digital vs Analog Time Conversion)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 4, Ch. 12 • CCSS.MATH.4.MD.A.2 • TEKS 4.8C

GRADE LEVEL BAND: Grades 2-5 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 6, pp. 152-158

HARDWARE UNBOXING BOOKLET: TIME-01 Booklet: Page 7 (RPN Stack Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

TIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE REVERSE ENCODING GAME

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Partner B inspects the physical clock hands and announces the time aloud.

Step 2: Partner B keys the time into the SC-6 calculator in CLK mode: e.g. [4] [:] [2] [5].

Step 3: Slide open the privacy shutter on the digital practice lid to verify the answer!

Step 4: Swap roles: Partner B sets the clock, Partner A reads and verifies.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Bimodal Duality: Physical Angular Displacement (Analog) = Numerical State Vector (Digital)

PHYSICAL APPARATUS MANIFEST:

Planetary Clock Core + Digital Practice Lid with Sliding Shu
tter

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

TIME-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE MULTI-STEP TIMELINE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

A student wakes up at 7:15 AM. Breakfast takes 25 min. Walking to school takes 15 min. When do they arrive?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Key start time into SC-6: [7] [:] [15] [ENTER].**
   Expected Physical Finding: **Y: 7:15 | X: 7:15**
2. **Add breakfast: [25] [+].**
   Expected Physical Finding: **X: 7:40**
3. **Add walk: [15] [+].**
   Expected Physical Finding: **X: 7:55 AM**
4. **Verify on the physical clock: Do the hands match 7:55?**
   Expected Physical Finding: **Minute hand at 11, hour pointer near 8**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [CLK] [8] [:] [45] [ENTER] [30] [+] -> 9:15

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
8:45	X: 8:45	Set initial analog clock time
ENTER	Y: 8:45 | X: 8:45	Stack lift
30	Y: 8:45 | X: 0:30	Key 30 minute delay
+	X: 9:15	Display updated time in register X

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack
spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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